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Attentional effects on affect and prefrontal activation during high-intensity exercise with music

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A R T I C L E I N F O

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A B S T R A C T

Music is widely used in exercise as a dissociative strategy to divert attention from sensations of fatigue. During high-intensity exercise, however, physiological perceptions may dominate attention, making it difficult for external stimuli like music to capture focus. Two experiments examined the impact of directing attention to music or body movements while running with asynchronous music. In Experiment 1, a within-subjects 2 (intensity: 70% or 80% heart rate reserve) x 2 (focus: music-focus or body-focus) design was employed. Focusing on music resulted in higher affective valence and remembered pleasure than focusing on body movements. No main effects of intensity level were observed. In Experiment 2, bilateral prefrontal oxygenation was measured using near-infrared spectroscopy during three conditions at 80%HRR: exercise with music and focus on music (music-focus), exercise with music and focus on body movements (body-focus), and exercise without music and focus on body movements (control). Compared to the control condition, the body-focus condition decreased prefrontal activation in the left hemisphere, suggesting that the presence of music reduced cognitive demands associated with regulating negative feelings during exertion. This reduction was not observed for the music-focus condition, possibly because actively processing the rhythmic structure of asynchronous music imposed additional cognitive demands. However, a mini-meta-analysis across both experiments revealed that focusing on music improved affective valence and remembered pleasure. There were also medium-sized positive correlations between perceived attention to music and affective responses. These findings suggest that attention to music is associated with positive affect even in high-intensity exercise where interoceptive information is dominant.

1. Introduction

Music is frequently used as an accompaniment to exercise. A meta-analysis of 139 empirical studies revealed that exercise with music significantly enhances affective valence and physical performance, while lowering perceived exertion and oxygen consumption, compared to exercise without music (Terry et al., 2020). One explanation is that music serves as a dissociative strategy because it distracts one's attention away from the internal sensations associated with physical exertion and fatigue during exercise (e.g., Karageorghis & Priest, 2012; Quan et al., 2025; Terry et al., 2020). The capacity of music to distract attention from exercise may be explained by the finite capacity for processing information in the central nervous system. During exercise, afferent feedback from working muscles travels from the body to the brain, where it directs attentional resources toward task-related information (i.e., associative thoughts) and contributes to a sense of fatigue or exertion (Pollak et al., 2014; Rejeski, 1985). The involvement of external stimuli like music reallocates attention toward information unrelated to the exercise task (i.e., dissociative thoughts), which inhibits the processing of physiological feedback signals and reduces awareness of fatigue-related sensations (Bigliassi et al., 2016, 2017, 2019).

However, this dissociative role of music to direct attention away from the sense of exertion is largely evident at low- to moderate-intensity exercise, but not during high-intensity exercise (Clark et al., 2016; Karageorghis & Priest, 2012). The parallel processing model proposes that, as sensory information is processed in parallel channels, individuals can intentionally shift their attention between interoceptive physiological signals (e.g., fatigue and discomfort) and external environmental cues such as music during low- or moderate-intensity exercise (Hutchinson & Tenenbaum, 2007; Rejeski, 1985). Conversely, during

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high-intensity exercise, peripheral signals about the body's internal state are most likely to dominate one's attention, reducing the capacity of music to inhibit these internal sensations (Hutchinson & Tenenbaum, 2007; Karageorghis & Priest, 2012; Patania et al., 2020; Rejeski, 1985). Indeed, as exercise intensity increases, there is a gradual shift from dissociative (e.g., mind-wandering) to associative (e.g., bodily sensations) thoughts (Hutchinson et al., 2015; Hutchinson & Tenenbaum, 2007; Jones et al., 2014; Karageorghis & Jones, 2013; Tammen & Tammen, 1996). Research suggests that music reduces ratings of perceived exertion (RPE) during moderate-intensity exercise to a greater extent than during high-intensity exercise, demonstrating that the beneficial effects of music is larger at moderate-intensity effort (Patania et al., 2020; Yamashita et al., 2006).

However, other evidence suggests that the benefits of music are not necessarily dependent on exercise intensity. A meta-analysis by Terry et al. (2020) showed that exercise intensity was not a significant moderator for the effects of music on RPE, affective valence, and physical performance. Findings consistently showed no significant differences in RPE and physiological measures (e.g., heart rate and blood lactate) between exercise with and without music during high-intensity exercise (Alloca Filho et al., 2022; Atan, 2013; Boutcher & Trenske, 1990; Hutchinson et al., 2011; Kemper, 2010; Maddigan et al., 2019; Tenenbaum et al., 2004; Yamashita et al., 2006). On the other hand, as an affective stimulus, music has been shown to enhance affective valence, enjoyment, and remembered pleasure toward the exercise, even during high-intensity exercise (Alloca Filho et al., 2022; Boutcher & Trenske, 1990; Hutchinson et al., 2011; Karageorghis et al., 2021). These findings for high-intensity exercise support the distinction between *what one feels* (i.e., fatigue sensation, as assessed by RPE) and *how one feels* (i.e., interpretation of the exertion, as assessed by measures such as affective valence), with the latter being more susceptible to environmental conditions such as music (Boutcher & Trenske, 1990; Hardy & Rejeski, 1989). In other words, although music may not significantly reduce the perception of fatigue under high-intensity workloads, it can color one's interpretation of fatigue, leading to a more positive evaluation (Karageorghis & Priest, 2012).

The extent to which individuals pay attention toward external perceptual input such as music (dissociative thoughts) plays a critical role in shaping affective responses during exercise (e.g., Bigliassi et al., 2019; Hutchinson et al., 2015; Jones et al., 2014). Hutchinson et al. (2015) found a significant correlation between dissociative thoughts and affective valence during exercise with music, irrespective of the intensity level relative to the ventilatory threshold (VT). Interestingly, the correlations were stronger at intensities above the VT, aligning with the finding by Jones et al. (2014). These findings suggest that even during high-intensity exercise, where attention is expected to be automatically dominated by associative thoughts, the more an exerciser is able to dissociate, the more likely they are to experience positive valence. However, this conclusion is based on self-reported levels of attention, rather than on manipulations that actively direct attention toward music during exercise.

There has been a longstanding focus on attention in sports and exercise research, as it plays a critical role in exercise performance and psycho/physical responses to exercise. There are multiple frameworks for characterizing attentional focus, including its direction (external vs. internal), breadth (broad vs. narrow), and task relevance (task-relevant vs. task-irrelevant) (see a review in Lind et al., 2009). In this study, we use the task-relevance approach introduced by Stevinson and Biddle (1998). According to this framework, attentional focus can be classified as association and dissociation, based on whether the focus is on the task. An associative strategy involves focusing on contents that are related to the exercise, like monitoring breathing or calculating mile splits. A dissociative strategy means focusing on non-exercise-related content, such as watching television, listening to music, or planning a vacation. Studies have shown that associative strategies decrease affective valence, whereas dissociative strategies facilitate exercise economy (e.g., Aghdaei et al., 2021; Hill et al., 2017; Lind et al., 2009). However, at high and near-maximal intensities, the advantages of dissociative strategies become smaller, likely because the overwhelming fatigue make it unavoidable to switch to association-dominant mode (Lind et al., 2009). Robertson and Marino (2016) suggested that focusing on motivational and emotional cues may modulate the tolerance of physiological sensations during high levels of physical exertion. Music is an external motivational and dissociative cue. It can disrupt awareness of the external world (Schäfer et al., 2013) and facilitate a flow state where movements feel effortless (Priest & Karageorghis, 2008), especially when music is absorbed into conscious experience (Howlin & Rooney, 2021).

A recent study using light-intensity isometric exercise has shown that focusing on music reduced RPE and enhanced affective valence compared to focusing on the feeling in the muscle (Moore et al., 2024). The difference in high-intensity exercise has not been investigated. Therefore, this study aims to compare the effects of a dissociative strategy of focusing on music and an associative strategy of focusing on body movement on RPE, affective outcomes (i.e., affective valence, perceived activation, and remembered pleasure), and bilateral prefrontal oxygenation during high-intensity exercise.

2. Experiment 1

Karageorghis and Jones (2013) found that 78% heart rate reserve (HRR) was a critical exercise intensity threshold at which attentional focus shifts from dissociative to associative with the presence of music. In Experiment 1, we investigated the effects of focusing attention on accompanying music versus body movements during high-intensity exercise at 70% and 80% HRR (i.e., either side of this critical intensity threshold). Music was present in both attentional allocation conditions, ensuring that any observed differences can be attributed to attentional allocation rather than confounded by the presence or absence of music. Asynchronous music, where music is not synchronous with movement, was used to isolate the role of attentional focus on music versus body movements during exercise. We hypothesized that explicitly instructing participants to focus on music would enhance its dissociative effect, resulting in lower RPE and more positive affective responses during exercise at 70% HRR. For exercise at 80% HRR, the prediction was less clear. On one hand, physiological sensations may dominate attention, potentially leading to null differences between the two attentional strategies. On the other hand, consciously directing attention toward music might amplify the dissociative influence of music accompaniment, promoting positive feelings toward the exercise compared to focusing on body movement.

3. Methods

The experiment was conducted using a 2×2 within-subjects design in which each participant engaged in four conditions involving 7-min treadmill runs. Two intensity conditions (70% HRR and 80% HRR) were crossed with two attentional allocation conditions (focus on music or body movement). The order of the four conditions was randomized and counterbalanced. All participants signed an informed consent form. This study was approved by the Macquarie University Human Ethics Committee (Reference No. 520231289053954).

3.1.1. Participants

G*Power 3.1 (Faul et al., 2009) was used to calculate the optimal sample size using the “Linear multiple regression: fixed model, R2 increase” statistical test. The parameters used were: power = 0.8, $p = .05$, $f = 0.31$ (transformed from partial $\eta^2 = 0.09$ as reported in Karageorghis & Jones, 2013), number of tested predictors = 1, and total number of predictors = 6.[1] It indicated that a minimum of 28 participants was required.

Participants were recruited through Sona System at Macquarie University, and external to the University through a flyer, Facebook, and RedNote (Xiaohongshu). The target population were young adults who regularly engage in recreational exercise. They were eligible to participate if they: 1) were between the ages of 18 and 35 years (Patania et al., 2020); 2) had no history of major health issues, including cardiovascular disease (e.g., heart attack and stroke), respiratory disease (e.g., asthma or chronic obstructive pulmonary disease), cardiopulmonary disease, metabolic disorders (e.g., diabetes), joint problems or musculoskeletal disorders, psychiatric or neurological disorders, or other conditions that prevent engagement in exercise; 3) had no history of alcohol or drug abuse; 4) were within normal weight ($18.5 < BMI < 25$) to minimize physiological variability (Di Lenarda et al., 2024) and potential confounding effects on the outcomes of interest such as perceived exertion (Ekkekakis & Lind, 2006); 5) were not pregnant or had recently given birth; 6) had regular physical activity for more than 75 min per week for the past six months; 7) had normal hearing; 8) were not professional athletes or professional musicians; 9) were classified as low risk via the Physical Activity Readiness Questionnaire (Warburton et al., 2011), with answering "no" to all seven questions.

Forty-four participants took part in the experiment. Four participants were excluded from the analysis: two did not complete all four conditions (one was intolerant of the exercise intensity and one had scheduling conflicts), one had a mean lower than 3 on the manipulation check question, and one had three outliers (i.e., $z > 3.29$) in the average heart rate (HR; indicating that the participant did not fall within the prescribed intensity range). Therefore, data from forty participants were included in the analysis (26 females, 13 males, and one non-binary). The mean age of the participants was 22.4 years ($SD = 4.5$). Their mean Body Mass Index was 21.49 kg/m^2. All participants reported engaging in aerobic exercise on a regular basis, with an average weekly duration of 240 min, assessed by the Personal Physical Exercise Habit Survey (Abramson et al., 2000). Thirty-two of them usually exercised with music.[2]

3.1.2. Measures

Participants rated their preferences and familiarity with a list of 13 motivational music pieces (Table S1) on a 7-point Likert Scale before participating in the onsite experiment. These music pieces, each with a tempo of 130 BPM, were selected for this experiment based on a pilot study (see Supplementary material). The tempo of 130 BPM was intentionally selected to prevent synchronization with the music, as the natural running cadence for most individuals typically falls between 160 and 180 steps per minute. Synchronization generally occurs when music tempo matches either one step per beat (i.e., 160–180 BPM) or two steps per beat (i.e., 80–90 BPM).

For the onsite experiment, participants rated the following measures immediately after each 7-min bout of running, without a gap in time between measures: (1) *Perception of the exercise exertion* was measured using Borg's RPE scale in response to the question "Please rate your perception of exertion" (Borg, 1998). As with other measures, exertion was not measured during exercise, as providing rating of any feelings would influence participants' attentional focus – a key variable in the study – to an associative focus (Hutchinson & Tenenbaum, 2007); (2) *Affective valence* was assessed using the Feeling Scale by responding to the item: "please rate your current feeling of mood from -5 (very bad) to 5 (very good)" (Hardy & Rejeski, 1989); (3) *Perceived activation* was assessed using the Felt Arousal Scale by responding to the item: "please rate your current feeling of arousal from 1 (low arousal) to 6 (high arousal)" (Svebak & Murgatroyd, 1985); (4) *Remembered pleasure* (i.e., how pleasant or unpleasant an event is remembered; see Hutchinson et al., 2018) was evaluated with an integer rating between −100 (very unpleasant) and 100 (very pleasant) in response to the question "How did the exercise session make you feel?" (Zenko et al., 2016; Zenko & Ladwig, 2021); (5) *Perceived attention toward music or physical effort* was assessed in two separate questions by answering "What percentage of time did you focus on the music / your physical effort", with options ranging from 0 (none of the time) to 100 (all the time); (6) *Post-exercise preference for the collection of music* selections in each condition was assessed by the question "how much did you like the music" on a scale from 1 (not at all) to 7 (strongly); (7) A *manipulation check* question was also included: "To what extent did you adhere to the instruction to focus on the music / your body movement[3]?" from 1 (not at all) to 7 (completely).

3.1.3. Procedure

A motorized treadmill (Nordic Fitness Trax Runner 2) was used for testing. Heart rate was collected using Polar H10. Music was played through JBL Endurance Run 2 earphones. Qualtrics was employed to counterbalance the conditions using the "loop and merge" design, display instructions, and administer measures. Participants attended the experiment over two visits, scheduled at the same time of day, with a minimum of 48 h and a maximum of one week between them. During the first visit, they completed a progressive maximal exercise test and two of the four experiment conditions, while the remaining two conditions were completed on the second visit. Participants were requested to 1) refrain from vigorous exercise for at least 24 h before the experiment; 2) avoid consuming caffeine and alcohol for at least 24 h before the experiment; 3) not eat within 2 h before the experiment; and 4) arrive hydrated. Upon arrival, participants were fitted with the Polar H10, which they wore throughout the visit.

On the first visit, participants were briefed on the experimental procedure. After sitting quietly in a comfortable chair and performing a 5-min box breathing while watching a guided meditation video, their resting heart rates (HR_{rest}) were recorded. This was followed by a 3-min warm-up on the treadmill at their comfortable speed. To determine their maximal heart rate (HR_{max}), participants completed a progressive maximal exercise test following the Bruce protocol (American College of Sports Medicine, 2017; Bruce & Hornsten, 1969). Given that all participants were regular exercisers, and thus had a higher baseline level of fitness compared to sedentary individuals, the protocol was adapted to begin at 5.5 km/h with an incline gradient of 14%. Each stage lasted three minutes. The test was terminated when any of the following criteria were met: 1) achievement of 95% of age-predicted peak HR (207–0.7*age); 2) an RPE of 19 or 20; 3) onset of angina or angina-like symptoms; 4) shortness of breath, wheezing, leg cramps, or claudication; 5) light-headedness, ataxia, nausea, or cold and clammy skin; 6) severe fatigue; 7) participant requests to stop. The mean of tested maximal heart rate was 184.08 BPM ($SD = 6.61$). To establish the exercise heart rate for the two intensity conditions, 70% and 80% of HRR were calculated for each participant using the Karvonen formula: Target HR $= [(HR_{max} - HR_{rest}) \times \%\ \text{intensity desired}] + HR_{rest}$.

Each condition began with 2–3 min without music, during which the researcher gradually adjusted the treadmill speed and incline gradient. The initial speed was set at 5.5 km/h, with a 1% gradient increase for

[1] The six predictors were exercise intensity condition, attentional allocation condition, music familiarity, music preference, presented order of conditions, and the random intercept for each participant. The tested predictor was either exercise intensity condition or attentional allocation condition.

[2] Linear mixed-effect models showed that having a habit of exercising with music did not significantly predict perceived activation ($\chi2(1) = 0.004$, $p = .95$), affective valence ($\chi2(1) = 0.24$, $p = .63$), remembered pleasure ($\chi2(1) = 0.15$, $p = .69$), RPE ($\chi2(1) = 1.9$, $p = .17$), perceived attention to music ($\chi2(1) = 0.29$, $p = .59$), nor perceived attention to physical effort ($\chi2(1) = 0.14$, $p = .71$).

[3] This prompt changed based on the condition.

every 0.5 km/h increment in speed (Karageorghis & Jones, 2013). A slight additional increase in gradient was implemented when participants' heart rates approached the target zone. The music began playing once participants reached the target heart rate. The exercise intensity continued at that workload for 7 min. The music tracks were randomly chosen from the list of motivational music without replacement, ensuring that the music selections in each condition were different for each participant. Participants rated the measures immediately following the 7-min run. The recovery period between each short bout of exercise lasted at least 5 min until participants' heart rates became stable.

For the two attentional allocation conditions, participants received different instructions. In the "focus on music" condition, participants were instructed: "Please focus your attention on music, and press the thumb counter every four beats". In the "focus on body movement" condition, the instruction was: "Please focus your attention on your body movement, and press the thumb counter every ten steps". An instruction "Please focus attention on music/body movement" was displayed on an iPad placed in the treadmill holder throughout the 7-min exercise. The manipulation of counting beats or steps was designed to ensure sustained engagement in the intended focus and to reduce potential attentional drift. Counting beats in music promotes prioritized attention to the metric framework, which can scaffold attentional resources and enhance - rather than detract from - the perception of other musical features, thereby supporting optimal allocation of attention during complex musical interactions (Keller, 1999; Keller & Burnham, 2005). Similarly, counting steps during running has been used as an associative strategy to focus on bodily function, as in the experiment of Aghdaei et al. (2021).

3.1.4. Data analysis

Linear mixed-effect models were performed to analyze the hypotheses using the *lme4* package (Bates et al., 2015) in R (R Core team, 2024). The model incorporated a random intercept for each participant to account for individual variability in baseline levels of the response variables. The fixed factor was Intensity × Attentional allocation. Music familiarity, collective post-exercise music preferences, and the presented order of conditions were included in the model as covariates to control their potential effects. Affective valence, perceived activation, remembered pleasure, RPE, perceived attention toward music or physical effort, and average and maximal HR were examined as response variables.

The significance of the main effects and interaction effects for the models were determined by the likelihood ratio test, where the models with and without the factor of interest were compared using the *anova()* function (Magezi, 2015; Winter, 2013). For example, the main effect for intensity was obtained by comparing the reduced model "dependent variable ~ attentional allocation + music familiarity + music preference + order + (1|participant)" with the model "dependent variable ~ intensity + attentional allocation + music familiarity + music preference + order + (1|participant)". Pairwise comparisons with Bonferroni adjustments were performed to identify significant differences between conditions using the *emmeans()* function. A p-value < .05 was considered significant for all analyses.

3.2. Results

Table 1 presents the means and standard deviations (SD) for each response variable under each condition. Diagnostic tests indicated that the assumptions of linear regression were met, except for affective valence which violated the normality assumption as revealed by the Shapiro test ($p = .03$). However, since both its histogram and Q-Q plot indicated that the residuals were approximately normally distributed, we did not transform the model of affective valence. Table 2 summarizes the main and interaction effects obtained from linear mixed-effect models. Fig. 1 illustrates the interaction between exercise intensity and attentional allocation for each response variable.

Table 1
Raw means and standard deviations in Experiment 1.

	70%HRR		80%HRR	
	Body	Music	Body	Music
Average HR	161.60 (8.80)	161.63 (9.10)	171.70 (9.57)	171.85 (9.09)
Maximal HR (task)	169.53 (9.35)	168.97 (9.62)	178.60 (10.57)	178.59 (9.60)
Perceived activation	3.88 (1.18)	4.05 (1.19)	4.08 (1.25)	4.28 (1.15)
Affective valence	2.27 (1.75)	2.87 (1.44)	2.08 (1.70)	3.15 (1.29)
Remembered pleasure	49.20 (32.95)	61.85 (28.07)	45.00 (37.68)	62.52 (26.98)
RPE	11.88 (2.04)	11.79 (1.59)	13.12 (2.11)	12.57 (1.84)
Perceived attention to music	42.67 (24.16)	78.03 (15.68)	41.02 (21.74)	71.90 (16.23)
Perceived attention to physical effort	60.00 (23.82)	34.77 (22.66)	65.20 (19.98)	42.55 (21.06)

Table 2
Likelihood ratio tests in Experiment 1.

	η_p^2	χ^2	p
Perceived activation			
Intensity	0.02	3.048	0.081
Attentional allocation	0.01	1.543	0.214
Intensity x Attentional allocation	0	0.407	0.523
Affective valence			
Intensity	0	0.044	0.835
Attentional allocation	0.07	8.837	0.003
Intensity x Attentional allocation	0	0.309	0.578
Remembered pleasure			
Intensity	0.01	0.601	0.438
Attentional allocation	0.08	9.656	0.002
Intensity x Attentional allocation	0	0.030	0.863
RPE			
Intensity	0.17	22.46	<0.001
Attentional allocation	0.01	0.873	0.35
Intensity x Attentional allocation	0.01	1.19	0.275
Perceived attention to music			
Intensity	0.02	2.038	0.153
Attentional allocation	0.47	79.303	<0.001
Intensity x Attentional allocation	0.02	2.09	0.148
Perceived attention to physical effort			
Intensity	0.04	4.627	0.031
Attentional allocation	0.30	44.28	<0.001
Intensity x Attentional allocation	0	0.386	0.535

Note. The models included music familiarity and music preference as covariates.

3.2.1. Manipulation checks

The manipulation checks for the average HR revealed no interaction ($\chi^2(1) = 0.29$, $p = .59$) and no main effect of attentional allocation ($\chi^2(1) = 0.45$, $p = .5$). However, a significant main effect of intensity was observed ($\chi^2(1) = 144.27$, $p < .001$), indicating a higher average HR at intensity of 80%HRR than 70%HRR ($t = 16.38$, 95%CI = [8.6, 10.96], $p < .001$). Similarly, no interaction ($\chi^2(1) = 0$, $p = .99$) nor main effect of attentional allocation ($\chi^2(1) = 1.37$, $p = .24$) was revealed for the maximal HR, but there was a significant main effect of intensity ($\chi^2(1) = 119.68$, $p < .001$). The pairwise comparison showed a significantly higher maximal HR at the intensity of 80%HRR than 70%HRR ($t = 14.03$, 95%CI = [7.91, 10.51], $p < .001$). The manipulation checks confirmed significantly higher average and maximal HR during intensity at 80%HRR than at 70%HRR, with no significant differences observed between the two attentional allocation conditions.

3.2.2. Affective outcomes

A significant main effect of attentional allocation on affective valence was found. Pairwise comparisons indicated that focusing on music had a significantly more positive affective valence than focusing

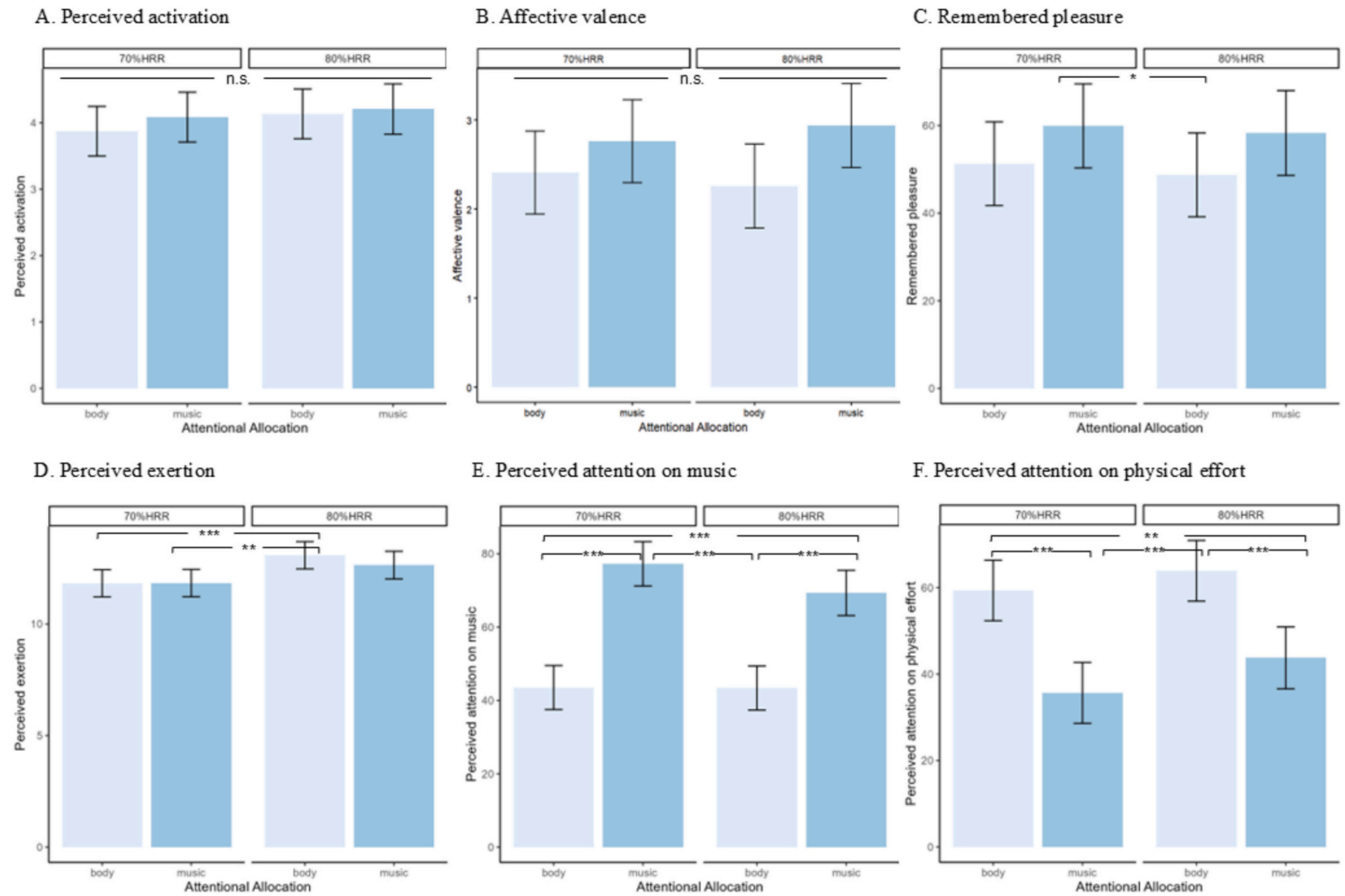

Fig. 1. Bar charts for the interaction between exercise intensity and attentional allocation in Experiment 1
Note. The values are the estimated marginal means obtained from the linear mixed-effect models that control music familiarity, music preference, and the order of conditions. These are the pairwise comparison results for the interaction effect. n.s. = non-significant; * $p < .05$, ** $p < .01$, *** $p < .001$.

on body movement ($t = 2.95$, 95%CI = [0.17, 0.86], $p = .004$). There was no significant interaction effect and no main effect of intensity on affective valence. For perceived activation, the main effect of intensity was marginally significant. There was no significant interaction effect and main effect of attentional allocation. For remembered pleasure, there was a significant main effect of attentional allocation. Pairwise comparisons indicated that focusing on body movement evoked lower remembered pleasure than focusing on music ($t = -3.07$, 95%CI = [−14.93, −3.23], $p = .003$). There was no significant main effect of intensity or interaction effect observed for remembered pleasure.

3.2.3. Perceived exertion

There was a significant main effect of intensity on RPE, with 80% HRR intensity having a significantly higher RPE than 70%HRR intensity ($t = 4.8$, 95%CI = [0.6, 1.46], $p < .001$). Interaction comparisons revealed that focusing on body movement at 80%HRR intensity was perceived as higher exertion than focusing on body movement ($t = 4.22$, 95%CI = [0.46, 2.05], $p < .001$) and music ($t = 4.04$, 95%CI = [0.42, 2.07], $p = .001$) at 70%HRR intensity. No other significant pairwise comparisons, main effect of attentional allocation, or interaction effect were observed.

3.2.4. Perceived attention

For perceived attention to music, there was a significant main effect for attentional allocation but not intensity. Pairwise comparisons indicated that participants spent more time focusing on music when they were in the music conditions ($t = 10.23$, 95%CI = [24.06, 35.61], $p < .001$). Regarding perceived attention to physical effort, there were significant main effects on both attentional allocation and intensity. Participants allocated more time focusing on their physical effort when they were in the body movement attentional allocation conditions ($t = 7.01$, 95%CI = [15.72, 28.1], $p < .001$). It was significantly higher at the intensity of 80%HRR than 70%HRR ($t = 2.13$, 95%CI = [0.43, 12.25], $p = .036$). There were no interaction effects for perceived attention to music and physical effort.

3.2.5. Associations among affective outcomes

The correlation matrix is shown in Table S2. There were significant correlations between perceived attention to music and perceived activation ($\rho = 0.16$, $p = .04$), affective valence ($\rho = 0.43$, $p < .001$), remembered pleasure ($\rho = 0.43$, $p < .001$), and RPE ($\rho = -0.29$, $p < .001$). Perceived attention to body movement significantly correlated with affective valence ($\rho = -0.19$, $p = .015$) and RPE ($\rho = 0.38$, $p < .001$). Linear mixed-effect models with perceived attention to music/body, exercise intensity, attention allocation, music preference, music familiarity, and order of conditions as the fixed factor, and participants as the random factor supported that perceived attention to music significantly predicted remembered pleasure ($\beta = 0.19$, $t = 2.11$, $p = .037$) and RPE ($\beta = -0.02$, $t = -3.16$, $p = .002$), but did not predict perceived activation ($\beta = -0.002$, $t = -0.47$, $p = .64$) and affective valence ($\beta = 0.009$, $t = 1.72$, $p = .087$). On the other hand, perceived attention to body movement significantly predicted RPE ($\beta = 0.02$, $t =$

4.1, $p < .001$) but not affective valence ($\beta = -0.009$, $t = -1.87$, $p = .06$).

3.3. Discussion

Participants reported greater perceived attention to music in the music-focus condition and greater attention to physical effort in the body-focus condition, confirming successful manipulation of attentional allocation. Under these conditions, our findings revealed that participants experienced significantly more positive affective valence and remembered pleasure when focusing on music compared to focusing on body movement, independent of the two exercise intensity levels. Unexpectedly, there were no main effects of intensity or interaction effects on any affective or psychophysiological responses. In other words, the difference between the two intensity levels – 70% and 80% HRR – was not large enough to produce distinctive effects. It should be noted that both intensities fall within the range of high-intensity exercise (60–89% HRR) according to the American College of Sports Medicine (2017) guideline. It is possible that both intensities exceeded the threshold at which physical stress dominates perception, such that feelings of fatigue were prominent regardless of these slight differences in intensity (Dyrlund & Wininger, 2008; Hutchinson & Tenenbaum, 2007; Karageorghis & Jones, 2013).

Although Karageorghis and Jones (2013) identified 78% HRR as a critical exercise threshold marking a shift from dissociative to associative focus when exercising with music, the small 2% difference between 78% (in Karageorghis & Jones, 2013) and 80% HRR (in the present study) may not reliably ensure that associative focus predominates at 80% HRR. To further evaluate the effect of exercise intensity on the shift from dissociative to associative focus, future research should compare intensity levels with larger differences (e.g., 65% vs. 85% HRR). Another limitation to consider is that participants completed the maximal exercise test and two experimental conditions during the first visit. Although the statistical analysis controlled the order of conditions, the exhaustive nature of the maximal exercise test might still have influenced performance in the subsequent conditions. Future research should separate the maximal exercise test from formal experimental sessions in different days. Overall, the results of Experiment 1 suggest that during high-intensity exercise, which is increasingly dominated by associative thoughts (Hutchinson & Tenenbaum, 2007; Tammen & Tammen, 1996), intentional attention toward music can still positively influence affective responses.

4. Experiment 2

Experiment 1 revealed significant differences between the two attentional allocation strategies on affective valence and remembered pleasure, confirming that there are benefits of attending to music while engaging in high-intensity exercise. However, self-reports are susceptible to demand characteristics and may not be sufficient to demonstrate that focusing on music prevents fatigue-related signals from entering focal awareness. To address this limitation, Experiment 2 employed a brain-imaging technique to reveal the brain mechanisms that underpin the effects of focusing on music in exercise. Combining self-reports with brain imaging should provide strong evidence for how music engagement can modulate fatigue perception in exercise.

Near-infrared spectroscopy (NIRS) is a non-invasive technique that detects tissue oxygenation and cerebrovascular responses in particular brain regions. It is relatively resistant to movement artifacts compared to other techniques (e.g., EEG and fMRI), and it also has an acceptable degree of spatial and temporal resolution, making it the most practical method for assessing neural activity during exercise (Karageorghis et al., 2018). To understand the top-down regulation, the prefrontal cortex (PFC) is our region of interest because it is strongly related to music-based emotion (Bigliassi et al., 2015). The PFC also serves numerous functions during exercise, including exercise tolerance (Hartman, 2020), intentional movement control (Lau et al., 2004), posture maintenance (Suzuki et al., 2004), emotion regulation and attention allocation (Bigliassi & Filho, 2022). It plays a crucial role in determining exercise tolerance by integrating uncomfortable physiological sensations with motivational and emotional cues from the environment (e.g., music) (Robertson & Marino, 2016).

Experiment 2 focused on the intensity level of 80%HRR to maximize the condition in which associative thoughts would be difficult to ignore. A no-music control condition was included to establish baseline data. During this control condition, participants were instructed to focus on body movement, allowing us to determine whether the mere presence of music can make a difference when attention allocation is held constant. Thus, there were three within-subjects conditions: focus on music, focus on body movement with music present, and focus on body movement with music absent (no-music control). All conditions were at 80% HRR. As prefrontal oxygenation increases and affective responses decrease with increasing exercise intensity (Tempest et al., 2014; Tempest & Parfitt, 2017), we hypothesized a negative association between activation in the PFC and affective outcomes in this high-intensity exercise paradigm. Given that Experiment 1 demonstrated that focusing on music enhanced affective valence and remembered pleasure, in this experiment we predicted that focusing on music would produce more positive affective responses and lower activation in both left and right PFC compared to focusing on the body and the no-music control.

4.1. Methods

4.1.1. Participants

The optimal sample size was calculated using G*Power 3.1 (Faul et al., 2009) with the "Linear multiple regression: fixed model, R2 increase" statistical test. A minimum of 11 participants was indicated after setting the following parameters: power = 0.8, α = 0.05, f = 1.11 (transformed from partial η^2 = 0.55 as reported in Yu et al., 2025), number of tested predictors = 1 (main effect of conditions), and total number of predictors = 3. Inclusion criteria used in Experiment 1 were employed, along with a requirement to be right-handed. This criterion was added to eliminate the potential impact of handedness on lateralization of brain function (see Costanzo et al., 2015; Dassonville et al., 1997). A total of 25 participants (7 males and 18 females) took part in this experiment. Their ages ranged from 22 to 35 years, with a mean of 28 years (SD = 4.14). Their mean Body Mass Index was 21.8 kg/m^2 (SD = 1.77). Twenty-four participants reported engaging in aerobic exercise on a regular basis, with an average weekly duration of 220 min. Fifteen of them usually exercised with music.

4.1.2. Music stimulus

Some participants in Experiment 1 reported that certain parts of some music tracks lacked clear audible beats, making it difficult to count beats accordingly. To address this limitation, we used the same music stimulus as in Bigliassi et al. (2018), "I Heard it Through the Grapevine" (119 BPM; 11:04 min; performed by Creedence Clearwater Revival; written by Barrett Strong and Norman Whitfield). This track has consistent audible beats and has a duration longer than seven minutes, allowing it to be used for the entire running bout without needing to switch tracks. Using a single and continuous piece of music helps avoid potential confounds due to musical characteristics that change across multiple tracks. The same track was used in both music conditions to minimize track variation (e.g., timbre, harmony, and motivational attributes). Audacity (2023) was used to normalize the perceived loudness to −23 loudness units full scale and adjust the tempo to 130 BPM.

4.1.3. Measures

All measures used in Experiment 1, including RPE, affective valence, perceived activation, remembered pleasure, and the manipulation check question, were also employed in this experiment. The perceived attention question was modified to: "What percentage of the time did you focus on 1) music (this item was not included in the no-music control condition); 2) other external thoughts (e.g., daydreaming, environment, distractions); and 3) internal body-related thoughts (e.g., heart rate, muscular fatigue, breathing)". The responses were not required to total 100%. At the end of the experiment, participants rated their preference and familiarity with the music. They also assessed the extent to which the piece of music would motivate them during exercise using the Brunel Music Rating Inventory (Karageorghis et al., 2006), since the motivational quality of music has been shown to influence in-task affect and endurance time (Elliott et al., 2005; Hutchinson & Karageorghis, 2013; Karageorghis et al., 2009).

4.1.4. NIRS data acquisition and analysis

Cerebral oxygenation was measured using two wireless continuous-wave NIRS devices (PortaLite, Artinis Medical Systems, Elst, The Netherlands). The PortaLite is a portable fNIRS system that generates fewer motion artifacts due to its lower inertia and allows more unrestricted movement than tethered systems, which are prone to wire-related optode shifts (Menant et al., 2020). It has three transmitters (i.e., LED optode sources) and one receiver, with transmitter-receiver distances of 30 mm, 35 mm, and 40 mm. Each transmitter emits near-infrared light with two wavelengths at 760 nm and 850 nm, primarily absorbed and reflected by deoxyhemoglobin (HHb) and oxyhemoglobin (O2Hb), respectively. The devices were placed at Fp1 and Fp2 according to the international 10–20 system, which roughly targets the left and right prefrontal cortex, respectively (Homan et al., 1987). Specifically, it was positioned at 10% of the nasion-inion distance from the nasion and 5% of the head circumference to the left and right from the midline. The devices were secured using double-sided adhesive tapes and protected from ambient light interference with a headband. Data was collected by OxySoft v3.5.15.4 and sampled with a frequency of 10 Hz. An age-based differential path length factor was used. Participants were asked to face forward and avoid sudden head movements, jaw clenching, frowning, or other facial expressions during the running bouts to minimize movement artifacts.

Raw NIRS data were pre-processed using the HOMER3 package (Huppert et al., 2009) in MATLAB v9.14 (The MathWorks Inc., 2023). Based on systematic comparisons among a range of motion artifact correction techniques, spline interpolation, wavelet analysis, the correlation-based signal improvement (CBSI) filter, and particularly the combination of wavelet and CBSI were favorable for minimizing the impact of motion artifacts (Al-Omairi et al., 2023; Brigadoi et al., 2014; Cooper et al., 2012; Tak & Ye, 2014). These filtering methods are recommended and widely used for reducing motion artifacts in tasks involving large movements (Herold et al., 2017; Menant et al., 2020). Therefore, these three techniques were employed in pre-processing. The specific processing streams are described as follows. First, the raw light intensity data were converted to optical density. Second, the SplineSG function was used to eliminate scalp movement noises by correcting baseline shifts using the spine interpolation method and correcting spikes using Savitzky-Golay filtering (Jahani et al., 2018). Third, wavelet filtering was employed to eliminate motion artifacts. Fourth, a fourth-order Butterworth bandpass filter with 0.01 to 0.5 Hz cutoff frequencies was used to remove cardiac-related high-frequency and low-frequency noises such as slow drifts and baseline wander (Herold et al., 2018). Fifth, optical densities were converted to concentration using the modified Beer-Lambert Law with a constant partial path-length factor of 1. Sixth, the CBSI filter was used to further remove motion artifacts and improve the signal. Finally, the block average was calculated to obtain O2Hb and HHb concentration for each condition and baseline.

The concentrations of O2Hb and HHb from the three transmitter-receiver pairs were averaged on the same side of the PFC to obtain a representative measure of activation, resulting in two values reflecting left and right PFC activity. The changes in oxyhemoglobin (ΔO2Hb), deoxyhemoglobin (ΔHHb), and hemoglobin difference (ΔHbDiff; HbDiff = O2Hb - HHb) were then calculated by subtracting the mean values during the 30-s resting baseline from the mean values during the conditions, which allows for assessing specific changes due to the exercise tasks.

4.1.5. Procedure

A motorized treadmill (Lifespan Fitness Viper M3) was used for testing. Heart rate was monitored via the Polar H10. Music was played through wireless over-ear sport earbuds (PocBuds V90). Qualtrics was employed to randomize the conditions, display instructions, and administer measures. The pre-experiment requirements for participants in Experiment 1 were also applied in this experiment.

Upon arrival, participants were fitted with the Polar H10 and PortaLite. Their HR_{rest} was recorded after they sat on a chair and performed 3-min box breathing while watching a guided meditation video to relax. Each condition began with a 30-s baseline period during which participants were required to stand on the treadmill, close their eyes, and count from one to 50 to prevent mind-wandering that may affect the recruitment of the prefrontal cortex (see Herold et al., 2018). Following this, there were 2–3 min of running without music to reach the targeted heart rate (i.e., 80% HRR). The Karvonen formula was used to calculate the HRR. The HR_{max} was determined by the age-predicted equation[4]: $HR_{max} = 192–0.007 \times age^2$ (Gellish et al., 2007), which is the most accurate among seven established equations (Cleary et al., 2011). The condition began immediately after the targeted heart rate was reached, at which point the PFC activity during running was recorded for seven minutes.

All three conditions were completed during a single visit, with their order randomized. For the two conditions involving exercise with music, participants were instructed to focus on either the music or their body movement by pressing a thumb counter every four beats or every ten steps, respectively. For the no-music control condition, no music was played during the running bout, and participants were instructed to focus on their body movement by pressing the thumb counter every ten steps. Participants rated the measures immediately after each condition. There was a recovery period of at least 5 min between each condition.

4.1.6. Data analysis

Linear mixed-effect models were conducted with the three conditions (i.e., music-focus, body-focus, control) as the fixed factor and participants as the random factor. The presented order of conditions was included as covariates to control for potential effects of repeated exposure to the same music. The response variables were average and maximal HR, affective responses (affective valence, perceived activation, and remembered pleasure), RPE, perceived attention, and prefrontal oxygenation (i.e., ΔO2Hb, ΔHHb, and ΔHbDiff) in the left and right prefrontal cortex. One-sample *t*-tests were conducted for ΔHbDiff to assess whether the change in neural activity significantly differed from zero, suggesting cerebral activation in the bilateral PFC. To control family-wise error across the two prefrontal regions of interest (left and right PFC), Bonferroni corrections were applied by multiplying each *p*-value by two. Spearman's correlations were used to explore associations between prefrontal oxygenation and affective responses due to non-normal distributions.

4.2. Results

Table 3 presents raw means and *SD*s for each response variable under each condition. Linear regression assumptions were met, except for the normality of perceived attention to internal thoughts ($p = .04$). However, since its Q-Q plot showed an approximately normal distribution, its model was not transformed. Table 4 summarizes pairwise comparisons and likelihood ratio tests.

[4] Using the age and HR_{max} data obtained in Experiment 1 ($N = 40$), the median difference between HR_{max} determined by the submaximal exercise test and HR_{max} calculated using the Gellish equation was 3.99.

Table 3
Raw means and standard deviations in Experiment 2.

	Control	Body-focus	Music-focus
Average HR	169.32 (5.63)	169.84 (4.61)	169.52 (5.38)
Maximal HR (task)	174.92 (6.12)	175.28 (5.44)	175.16 (5.52)
Perceived activation	3.36 (1.11)	3.96 (0.89)	3.96 (1.21)
Affective valence	1.92 (1.61)	2.40 (1.26)	2.40 (1.47)
Remembered pleasure	49.84 (29.8)	57.60 (22.93)	58.72 (25.24)
Perceived exertion	12.40 (1.76)	12.28 (1.21)	12.36 (1.66)
Perceived attention to external thoughts	54.80 (28.95)	36.23 (27.71)	30.00 (20.79)
Perceived attention to music	/	34.46 (21.64)	69.52 (23.47)
Perceived attention to internal thoughts	54.12 (28.14)	46.72 (24.67)	25.00 (19.33)
HbO – left PFC /10^{-3}	6.09 (10.65)	−2.74 (11.96)	3.90 (10.20)
HbR – left PFC /10^{-3}	−0.01 (0.01)	0.00 (0.01)	−0.00 (0.01)
HbDiff – left PFC /10^{-3}	12.28 (21.75)	−5.67 (24.36)	7.91 (20.97)
HbO – right PFC /10^{-3}	−4.09 (6.86)	−2.13 (11.23)	−3.31 (9.35)
HbR – right PFC /10^{-3}	0.00 (0.01)	0.00 (0.01)	0.00 (0.01)
HbDiff – right PFC /10^{-3}	−8.53 (14.26)	−4.69 (23.32)	−6.88 (19.25)

Table 4
Pairwise comparisons and likelihood ratio tests for each response variable in Experiment 2.

	Comparison	Estimate	95%CI	SE	t	p	Likelihood ratio test	η_p^2
Perceived activation	Control - body	−0.57	[−1.06, −0.07]	0.2	−2.84	0.02	$\chi^2(2) = 10.19, p = .006$	0.18
	Control - music	−0.55	[−1.05, −0.05]	0.2	−2.73	0.027		
	Body - music	0.02	[−0.48, 0.52]	0.2	0.1	1		
Affective valence	Control - body	−0.4	[−1.13, 0.34]	0.3	−1.34	0.563	$\chi^2(2) = 2.27, p = .32$	0.04
	Control - music	−0.35	[−1.09, 0.39]	0.3	−1.17	0.74		
	Body - music	0.05	[−0.7, 0.79]	0.3	0.16	1		
Remembered pleasure	Control - body	−6.92	[−16.77, 2.93]	3.96	−1.75	0.26	$\chi^2(2) = 4.52, p = .1$	0.09
	Control - music	−7.39	[−17.28, 2.51]	3.98	−1.85	0.21		
	Body - music	−0.46	[−10.41, 9.48]	4.00	−0.12	1		
RPE	Control - body	0.12	[−0.68, 0.92]	0.32	0.36	1	$\chi^2(2) = 0.15, p = .93$	0
	Control - music	0.04	[−0.77, 0.84]	0.32	0.12	1		
	Body - music	−0.08	[−0.89, 0.73]	0.33	−0.24	1		
ΔO2Hb – left PFC	Control - body	9.42	[1.51, 17.34]	3.17	2.97	0.03	$\chi^2(2) = 9.42, p = .009$	0.18
	Control - music	2.25	[−6.02, 10.51]	3.31	0.68	1		
	Body - music	−7.18	[−15.44, 1.09]	3.32	−2.16	0.22		
ΔHHb – left PFC	Control - body	−0.01	[−0.02, −0.00]	0.00	−2.95	0.032	$\chi^2(2) = 9.31, p = .01$	0.18
	Control - music	0.00	[−0.01, 0.01]	0.00	−0.64	1		
	Body - music	0.01	[−0.00, 0.02]	0.00	2.17	0.22		
ΔHbDiff – left PFC	Control - body	19.17	[3.00, 35.34]	6.48	2.96	0.03	$\chi^2(2) = 9.36, p = .009$	0.18
	Control - music	4.48	[−12.41, 21.37]	6.77	0.66	1		
	Body - music	−14.69	[−31.58, 2.20]	6.78	−2.17	0.22		
ΔO2Hb – right PFC	Control - body	−2.18	[−8.84, 4.49]	2.68	−0.24	1	$\chi^2(2) = 0.998, p = .61$	0.01
	Control - music	−2.38	[−9.23, 4.47]	2.75	−0.81	1		
	Body - music	−0.20	[−6.78, 6.37]	2.64	−0.87	1		
ΔHHb – right PFC	Control - body	0.00	[−0.01, 0.01]	0.00	0.73	1	$\chi^2(2) = 0.91, p = .63$	0.01
	Control - music	0.00	[−0.00, 0.01]	0.00	0.86	1		
	Body - music	0.00	[−0.01, 0.01]	0.00	0.16	1		
ΔHbDiff – right PFC	Control - body	−4.28	[−18.09, 9.53]	5.54	−0.77	1	$\chi^2(2) = 0.95, p = .62$	0.01
	Control - music	−4.92	[−19.11, 9.26]	5.69	−0.87	1		
	Body - music	−0.65	[−14.26, 12.97]	5.46	−0.12	1		
Perceived attention to external thoughts	Control - body	16.53	[0.69, 32.38]	6.36	2.6	0.038	$\chi^2(2) = 14.83, p = .001$	0.27
	Control - music	24.23	[8.78, 39.67]	6.20	3.91	0.001		
	Body - music	7.69	[−8.35, 23.73]	6.44	1.19	0.717		
Perceived attention to music	Body - music	−33.46	[−44.31, −22.62]	5.23	−6.39	<0.001	$\chi^2(1) = 26.61, p < .001$	0.64
Perceived attention to internal thoughts	Control - body	6.88	[−6.62, 20.37]	5.43	1.27	0.635	$\chi^2(2) = 25.04, p < .001$	0.39
	Control - music	28.67	[15.11, 42.24]	5.46	5.25	<0.001		
	Body - music	21.8	[8.17, 35.43]	5.48	3.97	<0.001		

4.2.1. *Manipulation checks*

There was no significant difference between conditions on average HR ($\chi^2(2) = 0.899$, $p = .64$) and maximal HR ($\chi^2(2) = 0.5$, $p = .78$), indicating that the intensity of exercise was the same across conditions. The average HR and maximal HR were 169.6 (5.16) BPM and 175.12 (5.63) BPM, respectively.

4.2.2. *Affective and psychophysical outcomes*

A significant main effect of perceived activation was observed. Pairwise comparisons showed lower perceived activation in the no-music control condition compared to both the body-focus ($t = -2.84$, 95%$CI = [-1.06, -0.07]$, $p = .02$) and music-focus conditions ($t = -2.73$, 95%$CI = [-1.05, -0.05]$, $p = .027$). No significant differences were found in affective valence, remembered pleasure, and RPE.

4.2.3. *Perceived attention*

Significant differences were observed between conditions in perceived attention to music, external thoughts, and internal thoughts. Participants spent more time paying attention to music in the music-focus condition than in the body-focus condition, indicating adherence to the instructions to allocate their attention. Compared to the control condition, both experimental conditions led to lower perceived attention to external thoughts. Perceived attention to internal thoughts was lower in the music-focus condition compared to both body-focus and control conditions, with no difference between body-focus and control conditions.

4.2.4. Prefrontal oxygenation

Of 75 measurements, seven from the left PFC and six from the right PFC were excluded because the data acquisition status values for all transmitters were close to 0%, indicating the tissue absorbed almost all the light. Additionally, one datum from the right PFC and one datum from the left PFC had z-scores larger than 3.29 and were also excluded from the analysis. On average, 0.6 channels were excluded per participant (range: 0–4 out of six), resulting in a mean retention rate of 90% across the sample. Therefore, 64 and 68 valid data for left and right PFC were analyzed, respectively. Fig. S4 illustrates the temporal dynamics of PFC oxygenation across the three experimental conditions for a participant.

The results for ΔHbDiff are reported below, as it serves as a sensitive indicator of regional cerebral blood flow and is commonly interpreted as reflecting cortical activation (Hoshi et al., 2001; Jones & Ekkekakis, 2019; Leff et al., 2011; Tsuji et al., 2000). The conclusions remained consistent when using ΔO2Hb and ΔHHb as outcome measures (see Table 4). In the control condition, there was significant activation in the left prefrontal region ($t(22) = 2.71$, 95%CI = [2.88, 21.69], $p = .026$), and significant deactivation in the right prefrontal region ($t(20) = -2.76$, 95%CI = [−14.98, −2.09], $p = .024$). No significant cortical activation was observed in the left PFC for both music-focus ($t(19) = 1.69$, 95%CI = [−1.91, 17.72], $p = .22$) and body-focus ($t(23) = -1.14$, 95%CI = [−15.95, 4.62], $p = .54$) conditions, nor in the right PFC for both music-focus ($t(22) = -1.72$, 95%CI = [−15.21, 1.44], $p = .2$) and body-focus ($t(23) = -0.99$, 95%CI = [−14.54, 5.16], $p = .68$) conditions. A significant main effect of the condition was observed in the left PFC but not in the right PFC. The ΔHbDiff of the left PFC in the body-focus condition was significantly lower than the control ($t = -2.96$, 95%CI = [3.02, 35.34], $p = .03$). No other significant differences were found.

4.2.5. Associations between prefrontal oxygenation and affective outcomes

The correlation matrix is shown in Table S3. Cerebral oxygenation showed no correlation with any affective responses. However, significant correlations were found between perceived attention to music and affective valence ($\rho = 0.34$, $p = .02$), remembered pleasure ($\rho = 0.31$, $p = .03$), and perceived attention to external thoughts ($\rho = -0.45$, $p = .002$). Linear mixed-effect models with perceived attention to music, condition, and the presented order of conditions as the fixed factor, and participants as the random factor supported that perceived attention to music significantly predicted affective valence ($\beta = 0.02$, $t = 2.18$, $p = .03$) and remembered pleasure ($\beta = 0.33$, $t = 2.58$, $p = .01$).

4.3. Discussion

The prefrontal oxygenation results revealed significant activation in the left PFC in the control condition, but no significant activation in either of the musical conditions. Previous studies have consistently shown that prefrontal activation increases with exercise intensity (Giles et al., 2014; Harada et al., 2009; Hartman, 2020; Jung et al., 2015; Leff et al., 2011; Rooks et al., 2010; Suzuki et al., 2004; Tempest & Parfitt, 2017; Thomas & Stephane, 2008), particularly in the ventral areas of the PFC (Tempest et al., 2014) that were examined in this experiment. Greater PFC activation during high-intensity exercise reflects a high cognitive demand required to maintain the exercise goal, inhibit the intensified interoceptive information from reaching the amygdala, and thus attenuate negative affective responses and modulate pain induced by fatigue (Ekkekakis, 2009; Harada et al., 2009; Hartman, 2020; Peng et al., 2018; Robert & Hockey, 1997; Tempest et al., 2014; Tempest & Parfitt, 2017). Based on this explanation, our findings suggest that the PFC activation in the no-music control condition likely indicates cognitive efforts to inhibit negative feelings. The PFC may have been activated to help participants tolerate exercise fatigue, protect the motor plan, and prevent task disruption in the absence of external distractions like music (Bigliassi & Filho, 2022). In contrast, the non-significant activation in both musical conditions may indicate that the presence of music, regardless of attentional focus, reduces the cognitive load on the PFC for emotion regulation. As Karageorghis et al. (2018) suggested, music can delay the increase in PFC oxygenation during exercise. Thus, music seems to reduce the need for high-level cognitive processing to modulate negative emotions and pain induced by high-intensity exercise. This result appears to contradict the hypofrontality theory, which posits reduced PFC activation and diminished cognitive performance during motor-cognitive dual-task conditions at high-intensity acute exercise, as cognitive resources are allocated to sensory and motor cortices (Dietrich & Audiffren, 2011; Jung et al., 2022). In contrast to cognitive tasks that demand substantial explicit processing such as short-term memory or executive function tasks, the present task involved directing attention to either music or bodily movement, which required relatively implicit processing and less reliance on prefrontal resources for task execution. Under this condition, PFC activation was upregulated to inhibit aversive sensations and maintain effort, rather than downregulated as predicted by hypofrontality.

Pairwise comparisons revealed that the left PFC was more activated in the control condition than in the body-focus condition, where the PFC showed a trend toward deactivation. Lower activation in the PFC suggests reduced reliance on cognitive resources, which indicates more efficient activation, better utilization of motor networks, and improved automaticity of movement (Maidan et al., 2018). As the only difference between the body-focus and control condition was the presence of music, background music may act as a distractor in the body-focus condition, reducing the demand for higher-level cognitive control and allowing participants to engage in more automatic processes, making the exercise feel less mentally taxing. Peng et al. (2018) also argued that distraction can reduce activation in many areas of the pain matrix and thus reduce pain perception.

However, contrary to our hypothesis, there was no significant difference in left PFC activation between the music-focus and control condition, nor between the music-focus and the body-focus condition. The music-focus condition showed a trend toward greater activation than the body-focus condition. Similarly, Odagiri et al. (2021) reported a non-significant trend of higher PFC oxygenation in the external focus condition than in the internal focus condition. In the music-focus condition, participants were asked to count musical beats that were not synchronized with their running pace. Some participants reported that this asynchrony was distracting, requiring them to shift attention between the music and the maintenance of running pace. In other words, actively focusing on the rhythmic structure of asynchronous music during exercise may have interfered with participants' optimal pacing strategy or diverted their attention from their preferred focus, as noted by Brick et al. (2020). Consequently, participants in the music-focus condition may have needed to exert greater cognitive effort to coordinate their leg movements and maintain a steady pace to avoid falling, such that the PFC oxygenation was not significantly reduced relative to the no-music control. In contrast, the body-focus condition required participants to focus on their pacing while music served as a background stimulus. This did not demand as much cognitive effort for attentional shifting.

In the right PFC, there was a significant deactivation in the control condition and non-significant deactivations in both musical conditions. This pattern of deactivation in the right PFC, coupled with activation in the left PFC in the control condition, suggests an asymmetry in PFC hemodynamics during high-intensity exercise. These findings align with previous research that showed greater left PFC activation compared to the right during exercise (e.g., Harada et al., 2009; Tempest et al., 2014). The left and right hemispheres of PFC may serve different functions in affective regulation during exercise. The left PFC is associated with approach-related behavior and the right with withdrawal-related behavior (Davidson, 1993; Harmon-Jones, 2004; Hartman, 2020; Tempest et al., 2014). Therefore, in the control condition, the asymmetric activation pattern – characterized by decreased right and increased left PFC oxygenation – may reflect reduced withdrawal-related processing (i.e., tendencies to disengage from the task or reduce effort), allowing approach-related behavior (i.e., sustained focus on the exercise task despite discomfort) to dominate cognitive resources. The bilateral PFC may work in tandem to support sustained engagement in high-intensity exercise. However, no significant difference between conditions was observed in the right PFC activation, suggesting that neither the presence of music nor the attentional allocation had a significant impact on modulating withdrawal-related behaviors during high-intensity.

Interestingly, there was no correlation between affective responses and hemoglobin difference in the bilateral PFC. This aligns with Tempest et al. (2014), who found weak non-significant correlations between affect and PFC activation at the ventilatory threshold, while a negative association emerged at higher intensities. They suggested that the sensory input from the body may not be strong enough to evoke negative emotions or compromise PFC capacity. Indeed, even in the control condition without music, participants' affective responses were relatively neutral (see Table 3). However, Tempest and Parfitt (2017) reported a positive correlation between prefrontal oxygenation and eyeblink amplitude (indicative of amygdala activity) at exercise intensity approaching the ventilatory threshold. The divergence between these two studies suggests that subjective self-reports and physiological signals may diverse, highlighting the importance of incorporating both subjective and objective measurements to more fully capture participants' affect.

Regarding subjective ratings, there was significantly more positive perceived activation in both musical conditions than in the control. However, no significant differences in affective outcomes were found between body-focus and music-focus conditions during high-intensity exercise. This finding contradicts the results of Experiment 1, where focusing on music led to significantly more positive affective valence and remembered pleasure compared to focusing on body movement. Importantly, the significantly higher perceived attention to music and lower perceived attention to internal thoughts in the music-focus condition confirm that the attentional manipulation was successful, so the non-significance was not attributed to ineffective manipulation. Also, both experiments had similar exercise intensities, as indicated by comparable average heart rates (171.77 BPM in Experiment 1 and 169.6 BPM in Experiment 2), ruling out intensity as a factor in the observed inconsistency. A key difference between the experiments was the selection of musical stimuli. In Experiment 1, three to four musical pieces were randomly selected from a list of motivational music (BMRI >31.5), while in Experiment 2, the same 7-min music track (median BMRI = 32) was used for both musical conditions. The correlational analysis (Table S3) revealed that higher BMRI scores were associated with higher affective valence and remembered pleasure. This finding suggests that the motivational quality of music may have influenced the emotional experience of exercise, which is consistent with previous research (e.g., Hutchinson & Karageorghis, 2013; Karageorghis et al., 2009). To address this potential confound, Experiment 2 used the same music track across conditions to ensure greater experimental control—allowing any observed effects to be more clearly linked to the attentional manipulation.

Nevertheless, the correlational analyses showed an important trend that participants' perceived attention to music was positively associated with affective valence and remembered pleasure. As Ekkekakis (2003) noted in the dual-mode model, neither interoceptive cues nor cognitive manipulations can fully control emotional responses to exercise. Indeed, participants in the music-focus condition did not devote their full attention to the music, with an average of 69.52% of their attention allocated to it (see Table 3). Given the high intensity of the exercise, it is inevitable that participants naturally divert some attention to sensations of physical fatigue. Likewise, even though participants were instructed to focus on their body movement in the body-focus condition, they still paid attention to music 34.46% of the time. The current correlational result suggests that the ability to focus more on music during high-intensity exercise can enhance positive affective experiences.

In conclusion, this experiment revealed two key findings for high-intensity exercise at 80%HRR. First, greater attentional focus on music was associated with more positive affective responses. Second, music accompaniment in exercise reduces the PFC's workload in regulating emotion. Specifically, significant activation in the left PFC during the no-music control condition likely reflects the brain's effort to attenuate negative feelings. In contrast, when music was present, regardless of attentional allocation, the PFC was not significantly activated. This suggests that music reduces the cognitive demand on the PFC, making the exercise less mentally demanding. A significantly lower left PFC activation in the body-focus music-present condition compared to the body-focus no-music control condition further supports the role of music in reducing reliance on cognitive resources for emotion regulation during high-intensity exercise. In contrast to the findings of Experiment 1, no significant differences were observed between music-focus and body-focus on affective valence and remembered pleasure, while greater perceived activation was observed in musical conditions than the no-music control in this experiment.

4.4. Mini meta-analyses

Given the divergent results of the two experiments regarding subjective affective ratings for the differences between music-focus and body-focus conditions at the intensity of 80%HRR, a mini meta-analysis of one's own results is recommended (see Goh et al., 2016; Mcshane & Böckenholt, 2017). A fixed-effect model was used because both experiments employed the same experimental manipulations and measurement scales. The *meatcont()* function in the *meta* package (Balduzzi et al., 2019) was used to pool effect sizes using Hedge's *g*. Across the two experiments, relative to focusing on body movement, focusing on music had small-to-medium positive effects on affective valence ($g = 0.42$ [0.07, 0.77], $\tau^2 = 0.18$, $I^2 = 73.2\%$) and remembered pleasure ($g = 0.34$ [−0.01, 0.69], $\tau^2 = 0.05$, $I^2 = 43.7\%$). Non-significant small effect sizes were found for perceived activation ($g = 0.1$ [−0.24, 0.45], $\tau^2 = 0$, $I^2 = 0\%$) and RPE ($g = -0.15$ [−0.49, 0.2], $\tau^2 = 0$, $I^2 = 0\%$).

To examine the correlations between perceived attention to music and affective and psychophysical outcomes at the intensity of 80%HRR, correlational meta-analyses using the *metacor*() function were conducted to pool effect sizes with Spearman's correlations. The results showed medium correlations between perceived attention to music and both affective valence ($\rho = 0.39$ [0.23, 0.53], $\tau^2 = 0$, $I^2 = 0\%$) and remembered pleasure ($\rho = 0.36$ [0.19, 0.5], $\tau^2 = 0$, $I^2 = 0\%$). There was also a significant small-to-medium negative correlation with RPE ($\rho = -0.22$ [−0.38, −0.05], $\tau^2 = 0.004$, $I^2 = 18.1\%$), and a marginally significant small-to-medium positive correlation with perceived activation ($\rho = 0.16$ [−0.02, 0.32], $\tau^2 = 0$, $I^2 = 0\%$). Perceived attention to body/internal thoughts was negatively correlated with affective valence ($\rho = -0.21$ [−0.37, −0.04], $\tau^2 = 0$, $I^2 = 0\%$) and positively correlated with RPE ($\rho = 0.28$ [0.11, 0.43], $\tau^2 = 0.12$, $I^2 = 87.8\%$), but showed no meaningful association with perceived activation ($\rho = 0.03$ [−0.14, 0.2], $\tau^2 = 0$, $I^2 = 0\%$) or remembered pleasure ($\rho = -0.05$ [−0.22, 0.13], $\tau^2 = 0$, $I^2 = 0\%$). These findings suggest that directing attention toward music during high-intensity exercise is associated with more positive affective responses and lower perceived exertion, while directing attention toward the body may have the opposite effect.

5. General discussion

The purpose of this study was to examine whether focusing externally on music or internally on body movement leads to different affective outcomes, RPE, and prefrontal oxygenation during high-intensity exercise. The mini meta-analysis showed that focusing on music led to higher affective valence and remembered pleasure compared to focusing on their body movements at the intensity of 80%HRR, as demonstrated in Experiment 1. Correlational analyses revealed a positive relationship between perceived attention to music and affective responses. Experiment 2 involved the measurement of prefrontal oxygenation and showed a significantly lower activation in the left PFC in the body-focus music-present condition than in the body-focus no-music condition, suggesting that music accompaniment in exercise supports more automatic movement processes in high-intensity exercise. No main effect of intensity levels in affective outcomes or main effect of attentional allocation in RPE was observed.

The dual-mode theory suggests that during high-intensity exercise, as intensity exceeds the body's capacity to maintain a physiological steady state, bottom-up interoceptive cues dominate attention and affective responses, making external cues or top-down cognitive factors (e.g., self-efficacy for exercise; shifts in attentional focus) difficult to influence affect (Ekkekakis, 2003, 2009; Hutchinson & Tenenbaum, 2007). Based on this model, it seems that cognitive manipulation would have minimal impact on affect during high-intensity exercise. However, our findings – primarily based on Experiment 1 and the mini meta-analysis – demonstrated that focusing on music significantly improved affective valence and remembered pleasure during high-intensity exercise, compared to focusing on body movement. Also, both experiments consistently showed that the more one was able to divert attention to music, the more positive affective responses they experienced during high-intensity exercise. These results suggest that top-down allocation of attention to external stimuli (e.g., music) can still positively influence affective responses, even in the face of the heightened salience of interoceptive information.

Consistently, both experiments did not find a difference in RPE between attentional allocation conditions. This aligns with previous research on high-intensity exercise with music (e.g., Alloca Filho et al., 2022; Hutchinson & Karageorghis, 2013; Karageorghis et al., 2013). RPE is influenced by the interaction of physiological, psychological, somatic and performance factors (Robertson et al., 2020). Consistent with the parallel processing model, this result suggests that during high-intensity exercise, RPE may be restricted to metabolic demands and less influenced by psychological factors such as music and attentional allocation (Brownley et al., 1995; Rejeski, 1985). Our findings suggest that although participants perceive exertion levels as similar across conditions, shifting attention from body movement to exercise-irrelevant stimuli (music) enhances their emotional interpretation of the exercise at high workloads.

Although this study highlights the importance of external focus on music during high-intensity exercise, there are several areas where future research can build upon and enhance these findings. First, assessing the ventilatory threshold, instead of heart rate reserve, would offer a more accurate measure of exercise intensity. Most studies determined the intensities of exercise using the ventilatory threshold and the respiratory compensation point. Even though Karageorghis and Jones (2013) used HRR and found a critical attentional shift threshold at 78% HRR, ventilatory threshold is more reliably related to changes in affect. Second, future studies can increase exercise intensity to the respiratory compensation point, to explore whether the external focus on music consistently influences affective responses and prefrontal oxygenation. Third, alternative methods of implementing an explicit focus on music can be considered, such as simply presenting a spoken instruction of "please focus on the music" every 15 s. In this study, participants were asked to count musical beats that were not in time with their running cadence. This asynchronization was designed on purpose, to ensure participants can focus on either body movements or music. However, this was sometimes distracting, requiring additional effort in movement coordination. Similarly, counting steps in the body-focus condition may also sometimes interfere with attending to physical sensations, lowering its strength of association. An alternative method such as presenting a spoken instruction of "please focus on your physical sensation" every 15 s can be considered in future research. Fourth, using a multi-channel fNIRS could provide more comprehensive insights into brain activity. Although the PortaLite was ideal for movement studies because of its lightweight and portable design, it measures only one to two regions at a time. Also, PFC oxygenation can reflect multiple processes (e.g., exercise tolerance, emotion regulation, and attention allocation) and it is affected by noise from body movements or surface blood flow. Careful steps were taken to minimize motion-related artifacts during running, yet residual artifacts cannot be entirely excluded and should be considered when interpreting the current neural activity results. A multi-channel fNIRS could assess a broader range of brain regions, offer richer data on brain function, and incorporate a separate channel to help regress out movement and superficial noise. Fifth, all outcomes were measured after the exercise in both experiments. While this approach was intended to minimize interference with attentional focus, it may have masked immediate responses experienced during the exercise. Lastly, this study targeted recreationally active young adults. Future studies could examine the effects in other populations (e.g., sedentary or elite athlete populations) to enhance generalizability.

In summary, this study provides empirical evidence for two key findings during high-intensity exercise: 1) the presence of music, when paired with a focus on body movement, reduces cognitive demand on the PFC needed to regulate emotions; and 2) focusing on music, while potentially more cognitively demanding due to the processing of asynchronous musical structure, enhances positive affective responses to exercise (i.e., affective valence and remembered pleasure). The hedonic responses may increase motivation to engage in future exercise sessions (Park et al., 2023). Indeed, a systematic review has shown that positive affective responses during exercise are associated with future physical activity engagement (Rhodes & Kates, 2015). Therefore, our findings encourage exercisers to listen to music and actively attend to it during high-intensity exercise. This can serve as a practical strategy to enhance immediate affective experience, boost motivation to continue exercise, and potentially improve long-term adherence to exercise routines.

CRediT authorship contribution statement

Yixue Quan: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Kirk N. Olsen:** Writing – review & editing, Supervision, Conceptualization. **William Forde Thompson:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Yixue Quan reports financial support was provided by China Scholarship Council. William Forde Thompson reports financial support was provided by Australian Research Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.actpsy.2026.107024.

Data availability

The data generated and/or analyzed during the current study are available from the corresponding author YQ on reasonable request.

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